

Full Stack + Cloud Engineering Roadmap (8-12 Months)

Goal: Become job-ready in Full Stack Development (Next.js, Node.js, NestJS) while building strong Cloud and DevOps skills (Linux, Docker, AWS, CI/CD) and continuing DSA preparation in Java.

Phase 1 (Weeks 1-4): JavaScript Fundamentals

- Variables, Data Types, Operators
- Conditionals and Loops
- Functions and Scope
- Arrays and Objects
- Mini Projects: Calculator, Number Guessing Game

Phase 2 (Weeks 5-6): Advanced JavaScript

- ES6+ Features
- DOM Manipulation
- Events
- Local Storage
- Promises
- Async/Await
- Fetch API

Phase 3 (Week 7): Git & GitHub

- Repositories
- Commits
- Branches
- Merge
- Pull Requests
- GitHub Workflow

Phase 4 (Week 8): TypeScript

- Types
- Interfaces
- Enums
- Generics
- Classes

Phase 5 (Weeks 9-12): React

- JSX
- Components
- Props
- State
- Hooks

- React Router
- Redux Toolkit

Phase 6 (Weeks 13-16): Next.js

- App Router
- Server Components
- Client Components
- Layouts
- Middleware
- Authentication

Phase 7 (Weeks 17-22): Backend Development

- Node.js Fundamentals
- Express.js
- REST APIs
- NestJS Modules, Controllers, Services, Guards

Phase 8 (Weeks 23-26): Databases

- MongoDB CRUD & Aggregation
- PostgreSQL
- Joins
- Indexes
- Transactions

Phase 9: Authentication & Security

- JWT
- Cookies
- OAuth
- Password Hashing
- RBAC

Phase 10: Linux & Networking

- Linux Commands
- Users & Permissions
- SSH
- DNS
- HTTP/HTTPS

- Ports

Phase 11: Deployment

- Ubuntu Server
- Nginx
- PM2
- SSL Certificates

Phase 12: Docker & CI/CD

- Docker Images
- Containers
- Docker Compose
- GitHub Actions Pipelines

Phase 13: AWS

- IAM
- EC2
- S3
- RDS
- VPC
- Load Balancers
- CloudWatch

Phase 14: Kubernetes & Terraform

- Pods
- Deployments
- Services
- Infrastructure as Code

Phase 15: System Design

- Caching
- Load Balancing
- Scalability
- Rate Limiting
- CDN
- Message Queues

Daily DSA Plan

- Arrays
- Strings
- Linked Lists
- Stacks
- Queues
- Trees
- Graphs
- Dynamic Programming
- Target: 300-500 LeetCode Problems

Capstone Projects

- LMS Platform
- Inventory Management System
- Conference Management System
- E-Commerce Platform
- Real-Time Chat App